

Literature Benchmark Brief: Nie_2023_ICLR

Title: A time series is worth 64 words: Long-term forecasting with PatchTST

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Cluster: Cluster E: Missing Data & Modular Risk Layers

Core Theoretical Mechanism & Model Formulation

Subseries patch extraction and channel-independent Transformer architecture

Empirical Scale & Experimental Dataset

Electricity, Weather, Traffic, and Exchange Rate benchmarks

Relevance to IEEE TSTE & Sustainable Energy Systems

Current SOTA long-sequence forecasting baseline

Identified Gaps & Operational Vulnerabilities

Channel independence discards cross-turbine spatial wake propagation

Role in Manuscript Defense & Reviewer #2 Proof

Synchronized baseline in Table I; demonstrates high RMSE in wind arrays